Computer Networks

Network Layer in Computer Networks

Unit - 6

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Store-and-Forward Packet Switching

Definition: A switching technique where a node (router) receives a full packet, stores it in a buffer, verifies the checksum for errors, and then forwards it to the next node.

Key Characteristic: The packet is not forwarded until it is completely received, allowing for error detection before transmission.

Network Layer

- The network layer in the TCP/IP protocol suite is responsible for the host-to-host delivery of datagrams.
- It provides services to the transport layer and receives services from the data- link layer.
- The network layer translates the logical addresses into physical addresses
- It determines the route from the source to the destination and also manages the traffic problems such as switching, routing and controls the congestion of data packets.
- The main role of the network layer is to move the packets from sending host to the receiving host.

Switching

Switching is one of the most important process in network World with **routing**. So, what is switching? **Switching** is the process of **forwarding** incoming packets from a port, to another port.

In **switching** there are two keywords for the ports used in this process. These are: **Ingress port** and **Egress port**. **Ingress port** is the port that the packet comes and the **egress port** is the port that the packet is forwarded.

Commonly, switching can be divided into two categories according to connection type. These are:

- **Connectionless**
- **Connection Oriented**

Switching

Connectionless Switching is the switching type that do not require any connection before the process. With the help of forwarding tables, switching process occurs.

Connection Oriented Switching is the switching type that requires a connection between two endpoints that the switching will be done. In other words, a circuit is established. The data is forwarded over this circuit in this type of switching.

Types of Switching

Different switching techniques can be used in different applications. These types are:

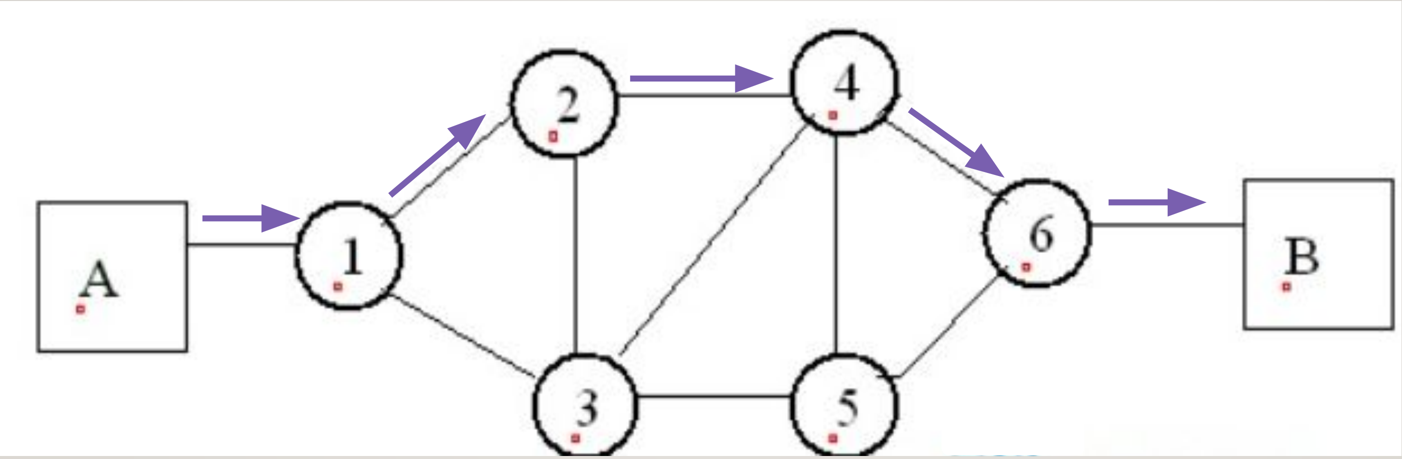
- Circuit Switching
- Packet Switching
- Message Switching

Circuit Switching

- Circuit switching is a switching technique that establishes a dedicated path between sender and receiver.
- In the Circuit Switching Technique, once the connection is established then the dedicated path will remain to exist until the connection is terminated.
- Circuit switching in a network operates in a similar way as the telephone works.
- A complete end-to-end path must exist before the communication takes place.
- In case of circuit switching technique, when any user wants to send the data, voice, video, a request signal is sent to the receiver then the receiver sends back the acknowledgment to ensure the availability of the dedicated path. After receiving the acknowledgment, dedicated path transfers the data.
- Circuit switching is used in public telephone network. It is used for voice transmission.
- Fixed data can be transferred at a time in circuit switching technology.

Circuit Switching

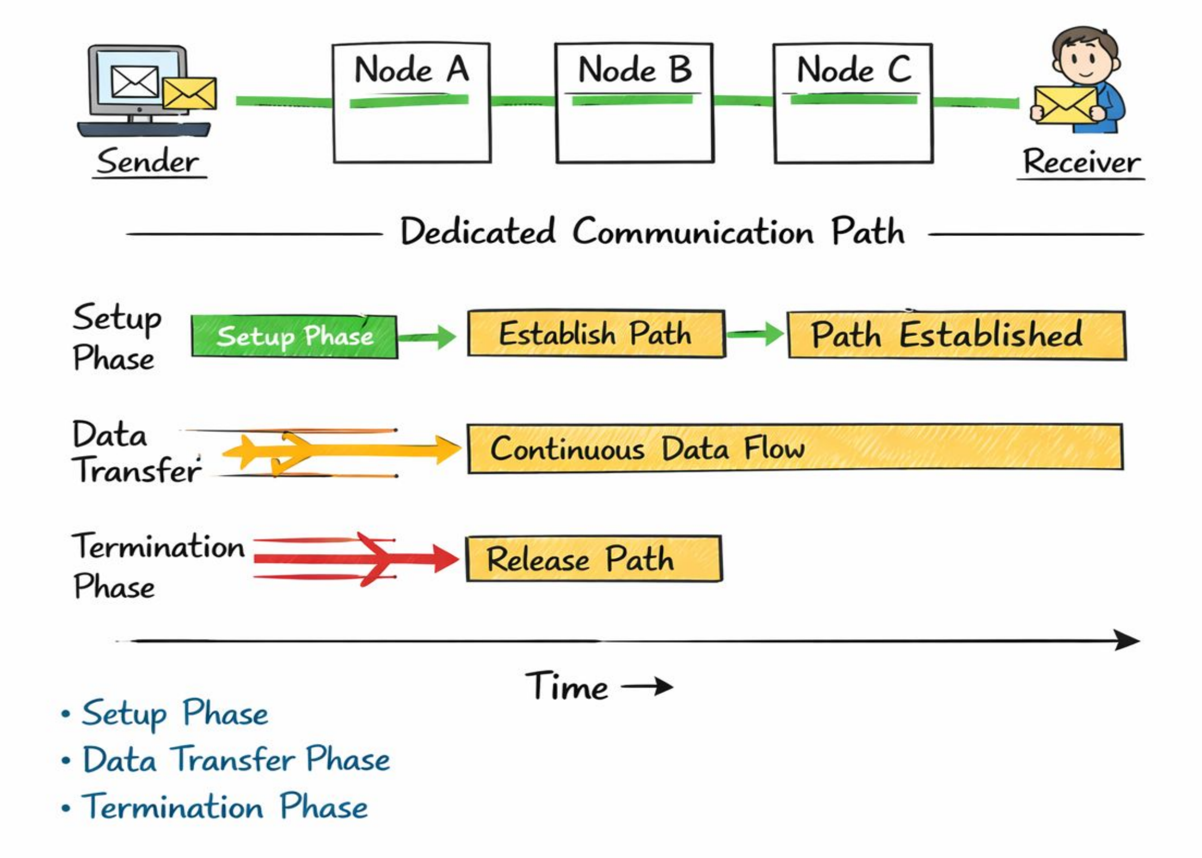
Source



Destination

Figure 1: Circuit Switching

Circuit Switching



Phases in Circuit Switching

Communication through circuit switching has 3 phases:

Connection Setup / Establishment - In this phase, a dedicated circuit is established from the source to the destination through a number of intermediate switching centres. The sender and receiver transmits communication signals to request and acknowledge establishment of circuits.

Data transfer - Once the circuit has been established, data and voice are transferred from the source to the destination. The dedicated connection remains as long as the end parties communicate.

Connection teardown / Termination - When data transfer is complete, the connection is relinquished. The disconnection is initiated by any one of the user. Disconnection involves removal of all intermediate links from the sender to the receiver.

Circuit Switching

Advantages of Circuit Switching

- It is suitable for long continuous transmission, since a continuous transmission route is established, that remains throughout the conversation.
- The dedicated path ensures a steady data rate of communication.
- No intermediate delays are found once the circuit is established. So, they are suitable for real time communication of both voice and data transmission.

Disadvantages of Circuit Switching

- In **Circuit Switching**, session establishment need for time so this causes more delays.
- The dedicated channel has advantages but they are expensive. So, this is an **expensive** solution. Beside this, if you do not use it enough, your resource can be wasted.
- This **type** works on **dedicated paths**. So, you can not use another data over this channel.

Packet Switching

- The packet switching is a switching technique in which the message is sent in one go, but it is divided into smaller pieces, and they are sent individually.
- The message splits into smaller pieces known as packets and packets are given a unique number to identify their order at the receiving end.
- Every packet contains some information in its headers such as source address, destination address and sequence number.
- Packets will travel across the network, taking the shortest path as possible.
- All the packets are reassembled at the receiving end in correct order.
- If any packet is missing or corrupted, then the message will be sent to resend the message.
- If the correct order of the packets is reached, then the acknowledgment message will be sent.

Packet Switching

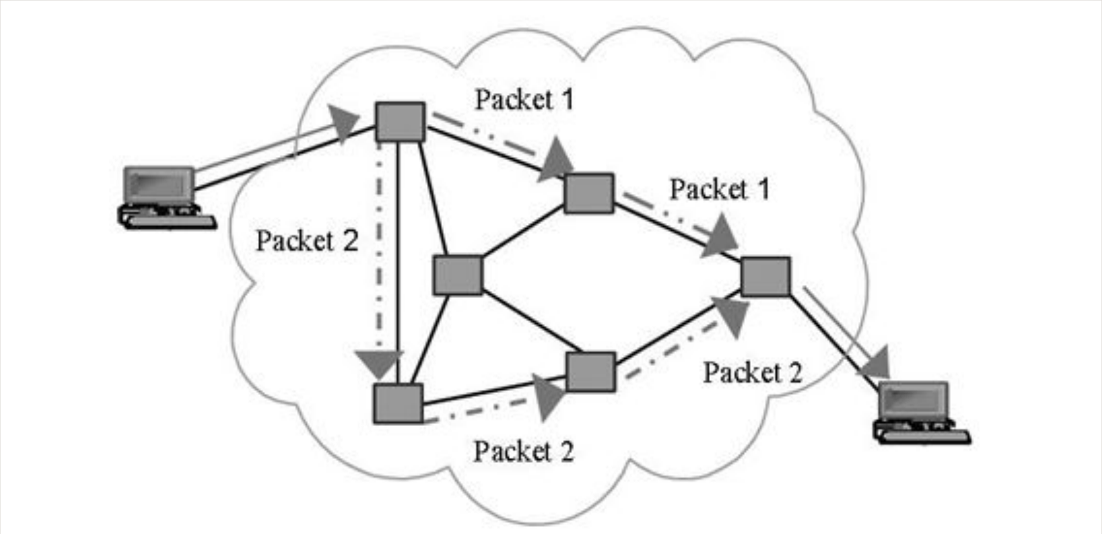
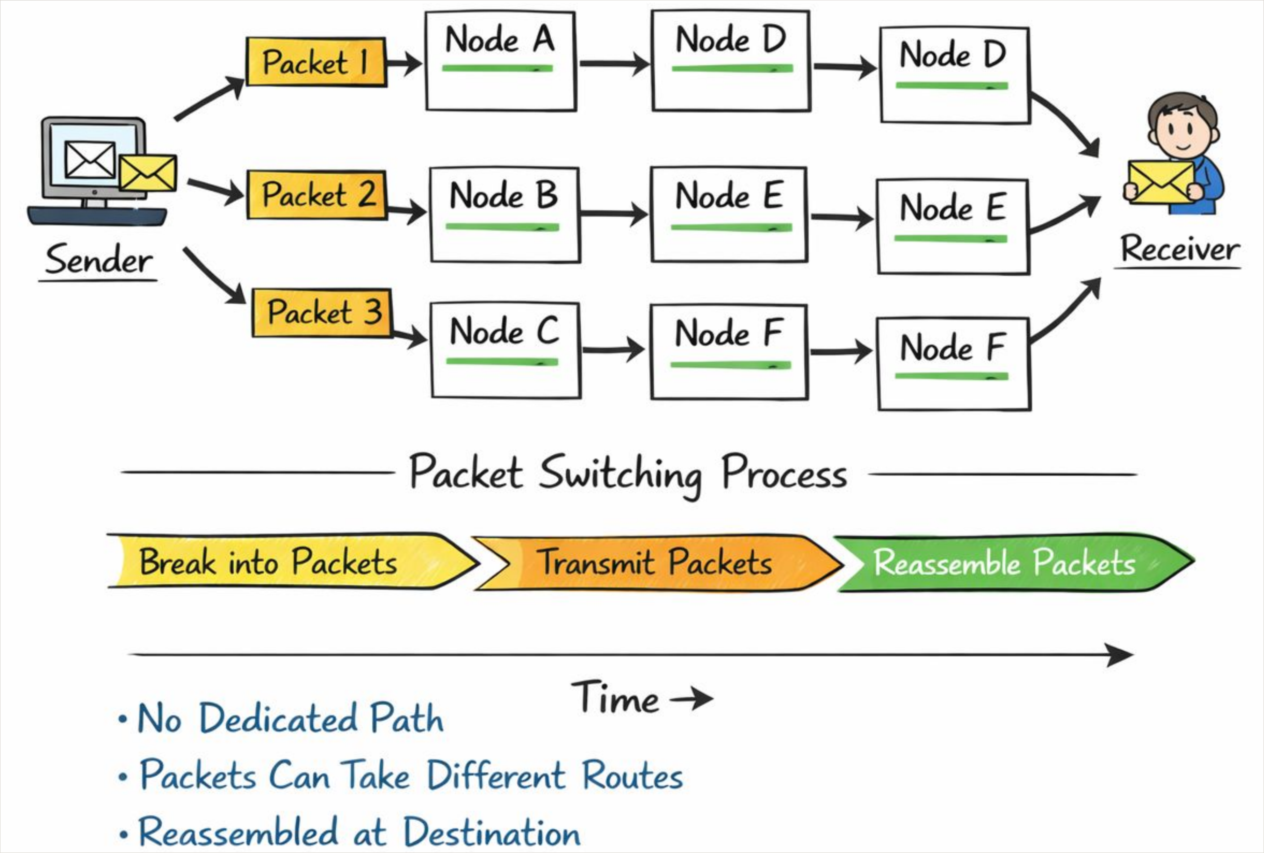


Figure 1: Circuit Switching

Packet Switching



Packet Switching

In packet switching, there are different techniques for data prioritization. Any data packet can be sent with a better priority with a better condition. Other packets can be sent over different priorities and conditions.

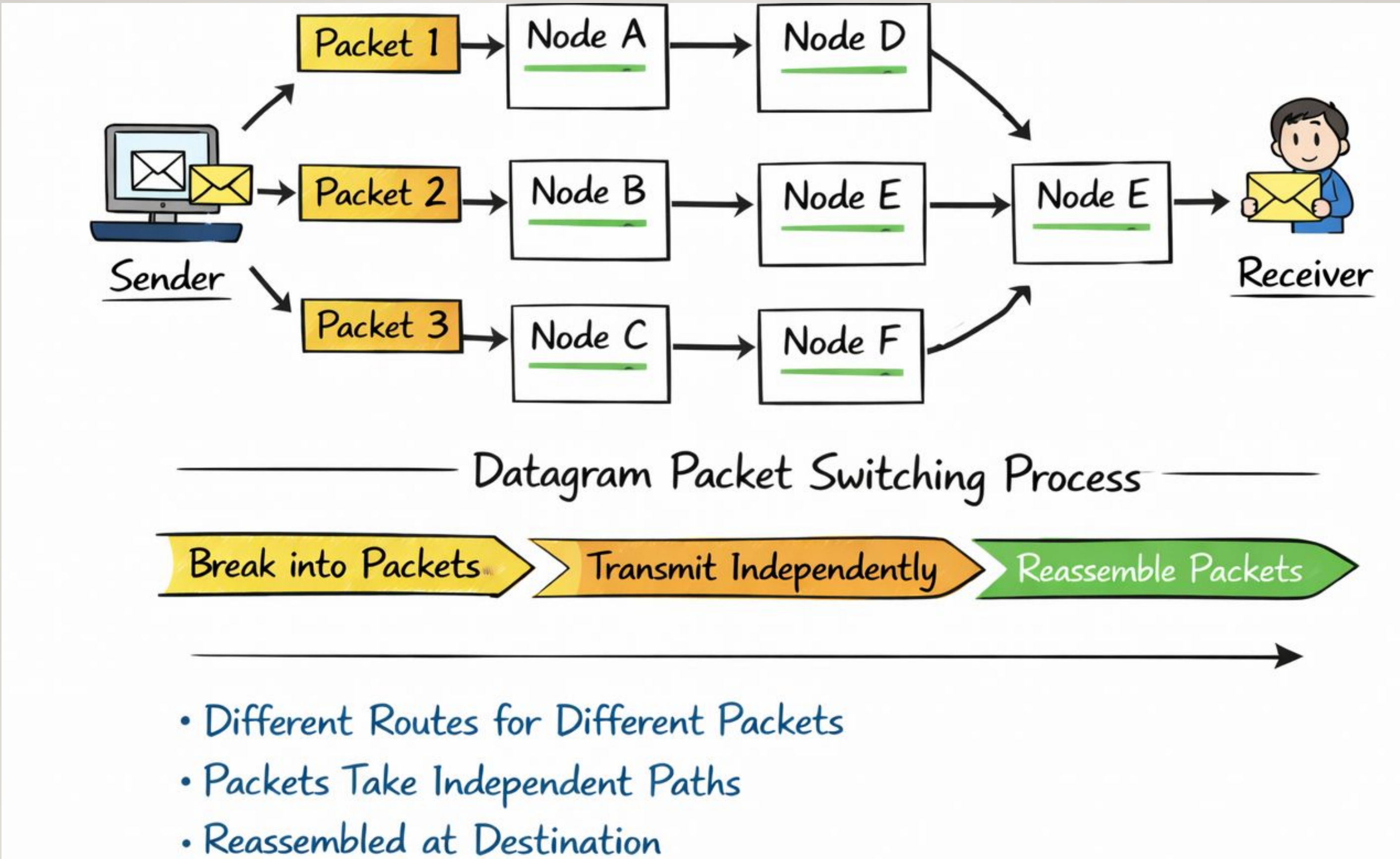
There are two approaches to Packet switching. These are:

- Datagram Packet
- Virtual Circuit

Datagram Packet Switching

- It is a packet switching technology in which packet is known as a datagram, is considered as an independent entity.
- Each packet contains the information about the destination and switch uses this information to forward the packet to the correct destination.
- The packets are reassembled at the receiving end in correct order.
- In Datagram Packet Switching technique, the path is not fixed.
- Intermediate nodes take the routing decisions to forward the packets.
- Datagram Packet Switching is also known as connectionless switching.
- There are no setup or teardown phases.
- Each packet is treated the same by a switch regardless of its source or destination.

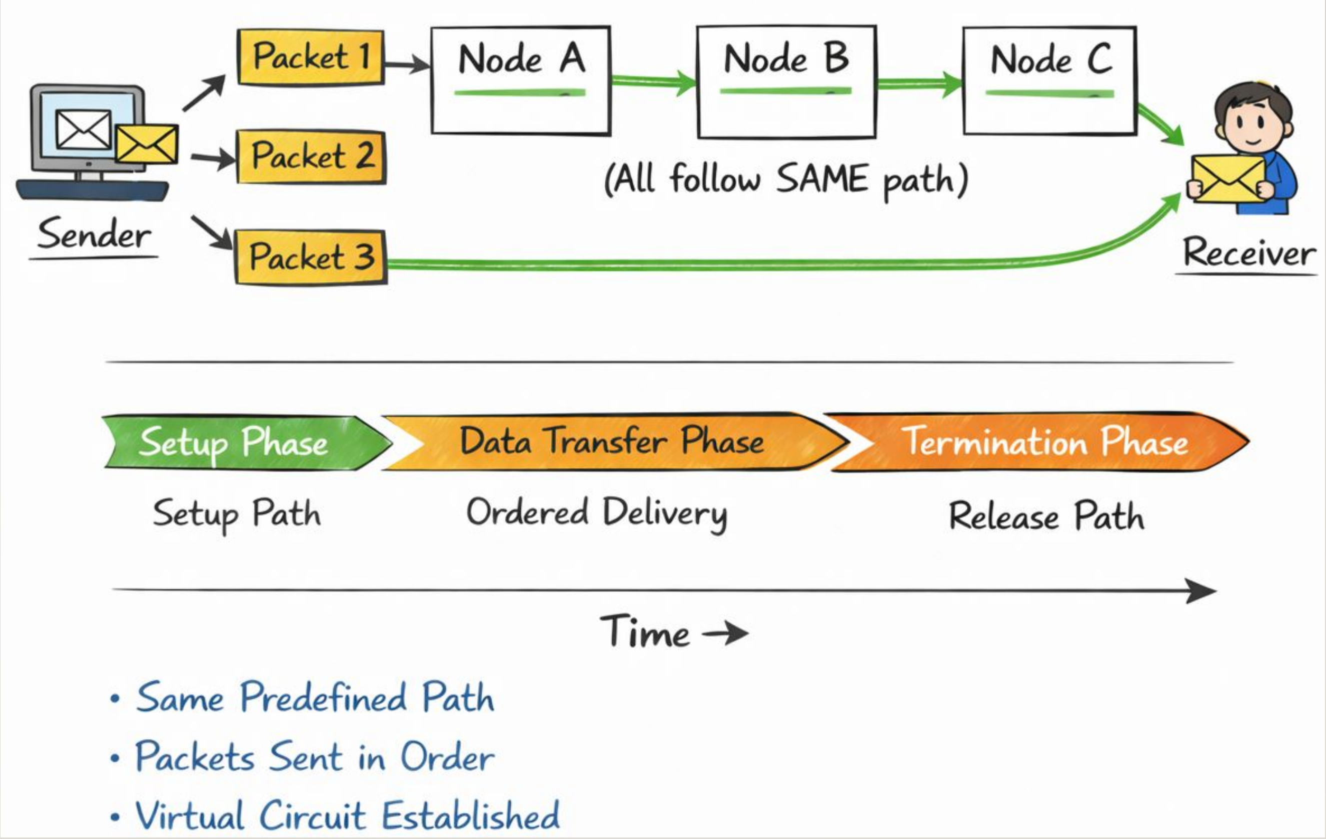
Datagram Packet Switching



Virtual Circuit Packet Switching

- Virtual Circuit Switching is also known as connection-oriented switching.
- In the case of Virtual circuit switching, a virtual connection is established before the messages are sent.
- Call request and call accept packets are used to establish the connection between sender and receiver.
- In this case, the path is fixed for the duration of a logical connection.

Virtual Circuit Packet Switching



Packet Switching

Advantages of Packet Switching

There are three main advantages of Packet Switching. These advantages are:

- Cost-effective
- Reliable
- Efficient

In packet switching, there is no need for store and forward mechanism. So, it need low resources. This make it a cost-effective solution.

The packet is sent another time if the destination is busy. So, this make packet switching a reliable mechanism.

By using different channels at the same time and there is no need for a fixed path, packet switching is an efficient mechanism.

Packet Switching

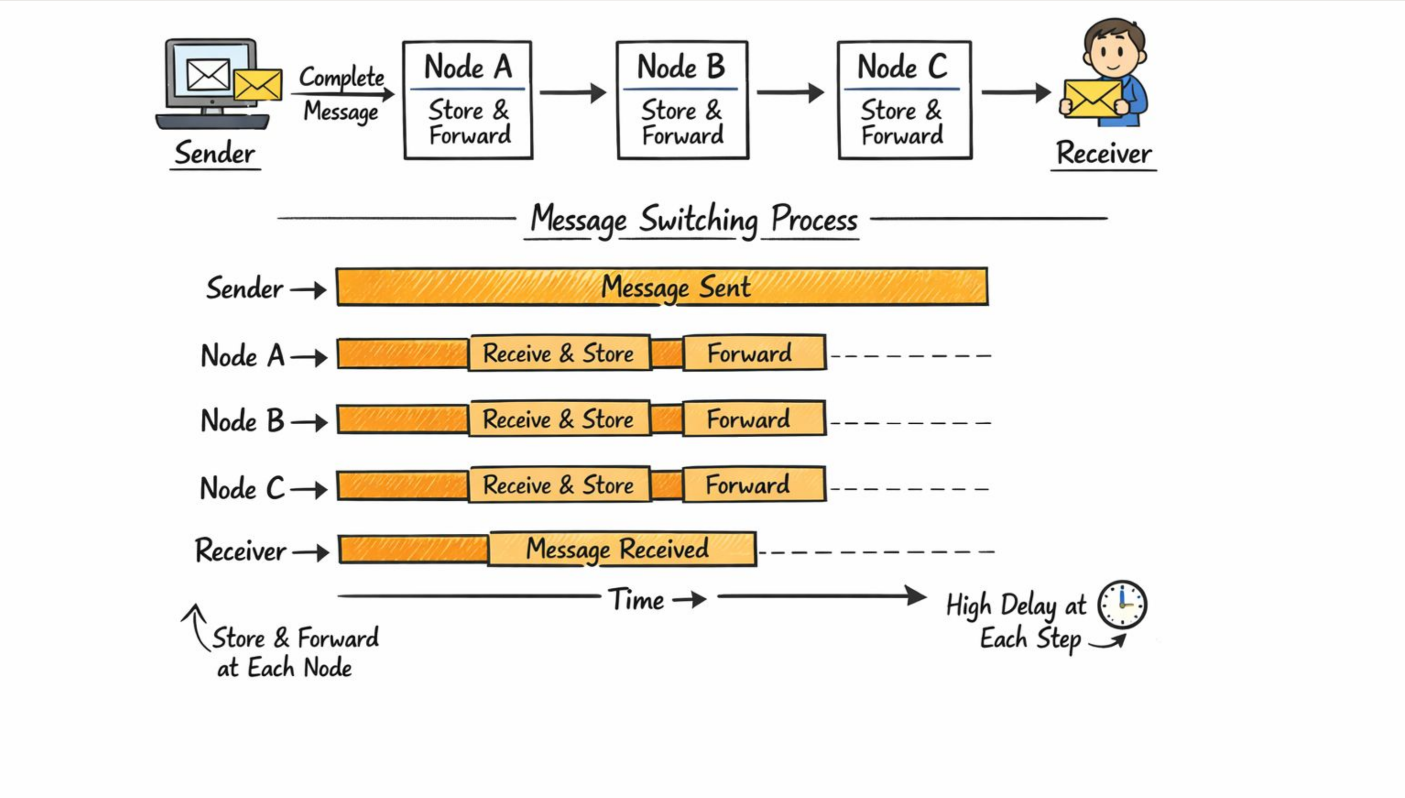
Disadvantages of Packet Switching

- Packet Switching technique cannot be implemented in those applications that require low delay and high-quality services.
- The protocols used in a packet switching technique are very complex and requires high implementation cost.
- If the network is overloaded or corrupted, then it requires retransmission of lost packets. It can also lead to the loss of critical information if errors are nor recovered.

Message Switching

- Message Switching is a switching technique in which a message is transferred as a complete unit and routed through intermediate nodes at which it is stored and forwarded.
- In Message Switching technique, there is no establishment of a dedicated path between the sender and receiver.
- The destination address is appended to the message. Message Switching provides a dynamic routing as the message is routed through the intermediate nodes based on the information available in the message.
- Message switches are programmed in such a way so that they can provide the most efficient routes.
- Each and every node stores the entire message and then forward it to the next node. This type of network is known as store and forward network.
- Message switching treats each message as an independent entity.

Message Switching



Message Switching

Advantages of Message Switching

- Message Switching is an efficient mechanism. Because, the data channels are shared between devices.
- Secondly, with Message Switching, traffic congestion is reduced. Because, there is a store and forward mechanism.
- The other advantage is, there is a prioritization mechanism in packet switching.
- The size of the message can be in different sizes in this type of switchig.

Message Switching

Disadvantages of Message Switching

The disadvantages of message switching are related to store and forward mechanism. This mechanism needs sufficient resources and it causes long delays. So, it is not suitable for the applications that need fast transmission. And this type of switching can be a little costly because of more resource requirement.

INTRODUCTION

- The Network Layer is the third layer of the OSI model.
- It handles the service requests from the transport layer and further forwards the service request to the data link layer.
- The network layer translates the logical addresses into physical addresses
- It determines the route from the source to the destination and also manages the traffic problems such as switching, routing and controls the congestion of data packets.
- The main role of the network layer is to move the packets from sending host to the receiving host.
- Unlike the transport layer (OSI Layer 4), which manages the data transport between the processes running on each host, network layer communication protocols (i.e., IPv4 and IPv6) specify the packet structure and processing used to carry the data from one host to another host.

SERVICES PROVIDED BY THE NETWORK LAYER

PACKETIZING

- The first duty of the network layer is definitely packetizing.
- This means encapsulating the payload (data received from upper layer) in a network-layer packet at the source and decapsulating the payload from the network-layer packet at the destination.
- The network layer is responsible for delivery of packets from a sender to a receiver without changing or using the contents.

SERVICES PROVIDED BY THE NETWORK LAYER

ROUTING AND FORWARDING

Routing

- The network layer is responsible for routing the packet from its source to the destination.
- The network layer is responsible for finding the best one among these possible routes.
- The network layer needs to have some specific strategies for defining the best route. Routing is the concept of applying strategies and running routing protocols to create the decision-making tables for each router.
- These tables are called as routing tables.

SERVICES PROVIDED BY THE NETWORK LAYER

ROUTING AND FORWARDING

Forwarding

- Forwarding can be defined as the action applied by each router when a packet arrives at one of its interfaces.
- The decision-making table, a router normally uses for applying this action is called the forwarding table.
- When a router receives a packet from one of its attached networks, it needs to forward the packet to another attached network.

SERVICES PROVIDED BY THE NETWORK LAYER

ERROR CONTROL

- The network layer in the Internet does not directly provide error control.
- It adds a checksum field to the datagram to control any corruption in the header, but not in the whole datagram.
- This checksum prevents any changes or corruptions in the header of the datagram.
- The Internet uses an auxiliary protocol called ICMP, that provides some kind of error control if the datagram is discarded or has some unknown information in the header.

SERVICES PROVIDED BY THE NETWORK LAYER

FLOW CONTROL

- Flow control regulates the amount of data a source can send without overwhelming the receiver.
- The network layer in the Internet, however, does not directly provide any flow control.
- The datagrams are sent by the sender when they are ready, without any attention to the readiness of the receiver.
- Flow control is provided for most of the upper-layer protocols that use the services of the network layer, so another level of flow control makes the network layer more complicated and the whole system less efficient.

SERVICES PROVIDED BY THE NETWORK LAYER

CONGESTION CONTROL

- Another issue in a network-layer protocol is congestion control.
- Congestion in the network layer is a situation in which too many datagrams are present in an area of the Internet.
- Congestion may occur if the number of datagrams sent by source computers is beyond the capacity of the network or routers.
- In this situation, some routers may drop some of the datagrams.

SECURITY

- Another issue related to communication at the network layer is security.
- To provide security for a connectionless network layer, we need to have another virtual level that changes the connectionless service to a connection-oriented service. This virtual layer is called as called IPSec (IP Security).

Virtual Circuit Switching	
Connection oriented	Connection less
Ensures in order delivery	Packets may be delivered out of order
No reordering is required	Reordering is required
A dedicated path exists for data transfer	No dedicated path exists for data transfer
All the packets take the same path	All the packets may not take the same path
Resources are allocated before data transfer Resources are allocated on demand using 1st packet No resources are allocated Stream oriented Packet oriented Packet oriented Fixed bandwidth Dynamic Bandwidth Dynamic bandwidth	Reliable Reliable Unreliable No overheads Less overheads Higher overheads Implemented at physical layer Implemented at data link layer Implemented at network layer Inefficient in terms of resource utilization Provides better efficiency than circuit switched systems Provides better efficiency than message switched systems Example- Telephone systems Examples- X.25, Frame relay Example- Internet

IP ENCAPSULATION

IP encapsulates the transport layer (the layer just above the network layer) segment or other data by adding an IP header. The IP header is used to deliver the packet to the destination host.

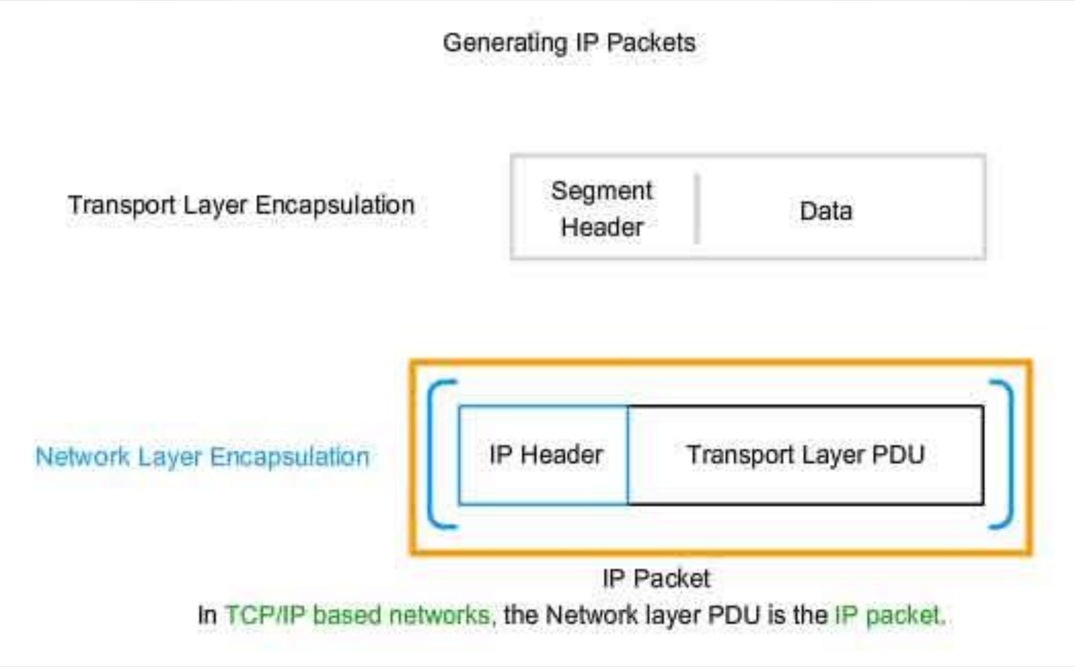


Figure 1: IP packet encapsulation from transport layer to network layer

CHARACTERISTICS OF IP

- IP was designed as a protocol with low overhead. It provides only the functions that are necessary to deliver a packet from a source to a destination over an interconnected system of networks. The protocol was not designed to track and manage the flow of packets.
- These are the basic characteristics of IP:
  i. **Connectionless** - There is no connection with the destination established before sending data packets.
  ii. **Best Effort** - IP is inherently unreliable because packet delivery is not guaranteed.
  iii. **Media Independent** - Operation is independent of the medium (i.e., copper, fiber-optic, or wireless) carrying the data.

IPv4 Communication

- The IPv4 protocol can be involved in three types of communication: unicasting, multicasting and broadcasting.
- **Unicasting** is the communication between one sender and one receiver.
  - It is a **one-to-one** communication.
- However, some processes sometimes need to send the same message to a large number of receivers simultaneously. This is called **multicasting,**
  - which is a **one-to-many** communication.
- Multicasting has many applications. For example, multiple stockbrokers can simultaneously be informed of changes in a stock price, or travel agents can be informed of a plane cancellation. Some other applications include distance learning and video-on-demand.
- **Broadcast** transmission refers to a device sending a message to all the devices on a network in one-to-all communications.
- IPv4 uses broadcast packets. There are no broadcast packets with IPv6.

Types of IPv4 addresses

1. **Public IPv4 addresses**
   - Public IPv4 addresses are addresses which are globally routed between internet service provider (ISP) routers.
2. **Private IPv4 addresses**
   - There are blocks of addresses called private addresses that are used by most organizations to assign IPv4 addresses to internal hosts. Private IPv4 addresses are not unique and can be used internally within any network.
3. **Special Addresses**
   - **Loopback addresses** (127.0.0.0/8 or 127.0.0.1 to 127.255.255.254) are more commonly identified as only 127.0.0.1, these are special addresses used by a host to direct traffic to itself.
   - **Link-local addresses** (169.254.0.0 /16 or 169.254.0.1 to 169.254.255.254) are more commonly known as the Automatic Private IP Addressing (APIPA) addresses or self-assigned addresses. They are used by a Windows DHCP client to self-configure in the event that there are no DHCP servers available.

IPv4-Classful Addressing

IPv4 addresses were assigned using classful addressing and customers were allocated a network address based on one of three classes, A, B, or C. The RFC divided the unicast ranges into specific classes as follows:

1. **Class A (0.0.0.0/8 to 127.0.0.0/8)** - Designed to support extremely large networks with more than 16 million host addresses. Class A used a fixed /8 prefix with the first octet to indicate the network address and the remaining three octets for host addresses (more than 16 million host addresses per network).

2. **Class B (128.0.0.0 /16 - 191.255.0.0 /16)** - Designed to support the needs of moderate to large size networks with up to approximately 65,000 host addresses. Class B used a fixed /16 prefix with the two high-order octets to indicate the network address and the remaining two octets for host addresses (more than 65,000 host addresses per network).

3. **Class C (192.0.0.0 /24 - 223.255.255.0 /24)** - Designed to support small networks with a maximum of 254 hosts. Class C used a fixed /24 prefix with the first three octets to indicate the network and the remaining octet for the host addresses (only 254 host addresses per network).

4. There is also a **Class D** multicast block consisting of 224.0.0.0 to 239.0.0.0 and a **Class E** experimental address block consisting of 240.0.0.0 - 255.0.0.0.

IPv4-Classful Addressing

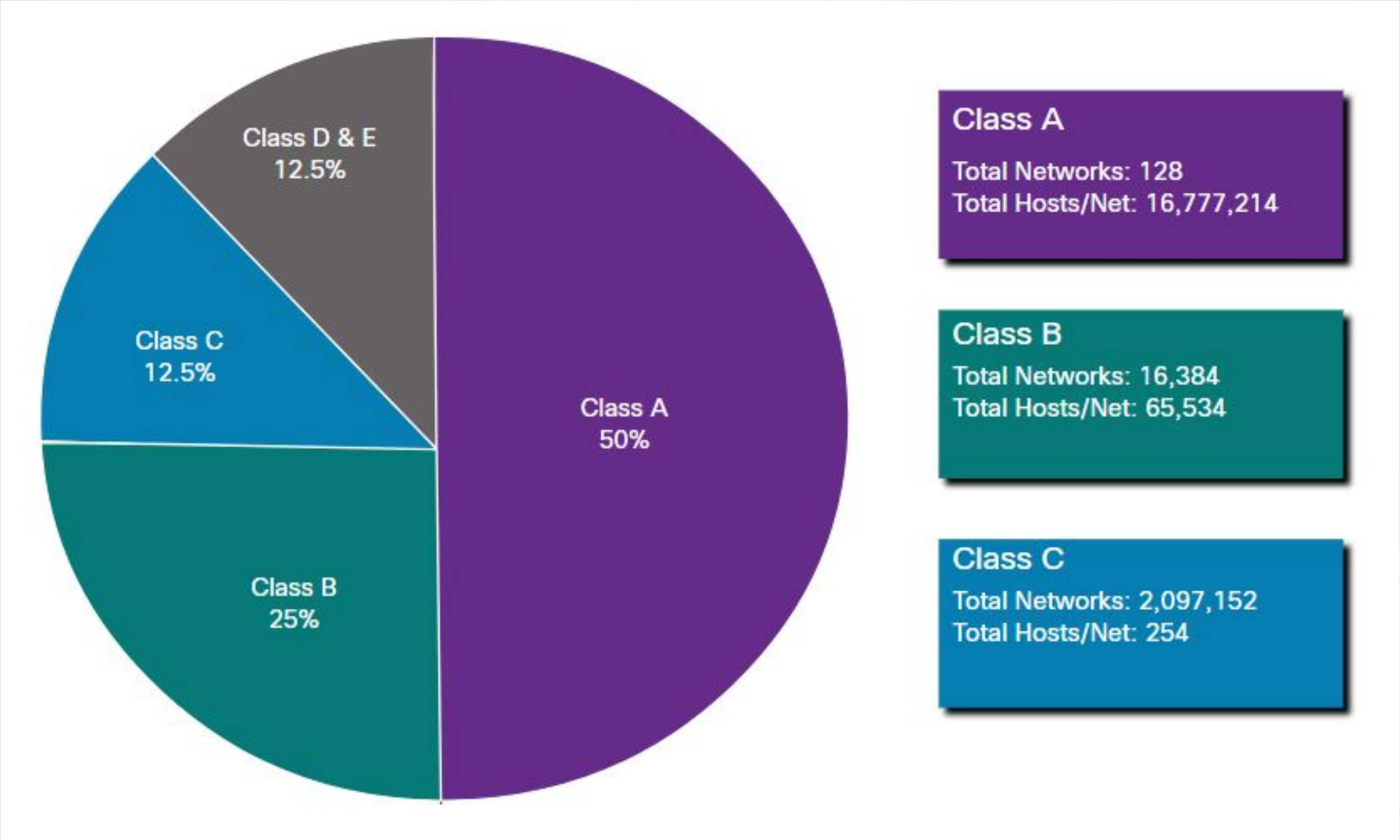


Figure: IPv4 Classful addressing

IPv6 Address

- IPv6 was developed by Internet Engineering Task Force (IETF) to deal with the problem of IPv4 exhaustion. IPv6 is a 128-bits address having an address space of 2^{128}, which is way bigger than IPv4. IPv6 use Hexa-Decimal format separated by colon (:) .

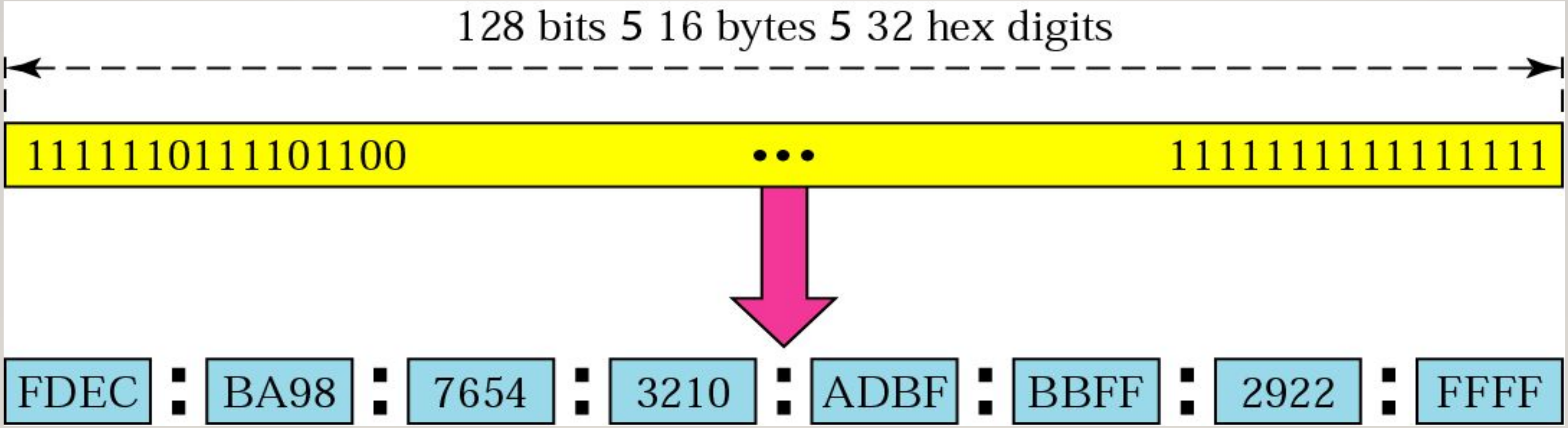


Figure IPv6 Address Format

ABBREVIATED ADDRESS

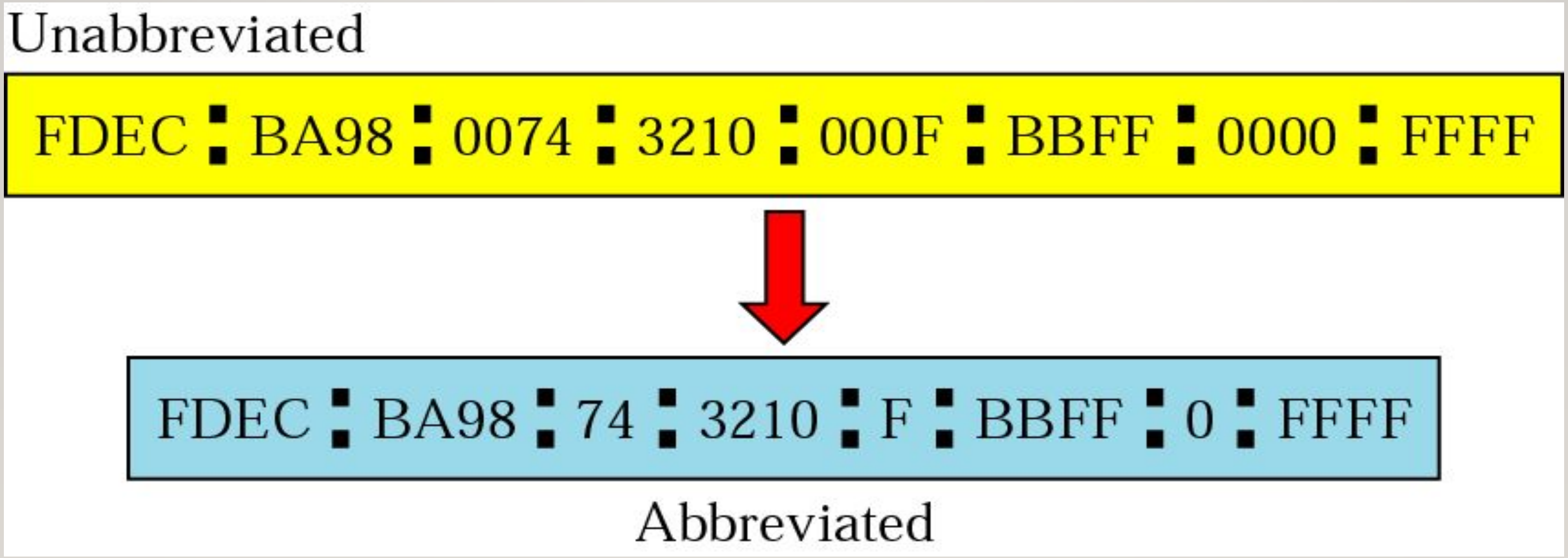


Figure IPv6 Abbreviated Address

ABBREVIATED ADDRESS WITH CONSECUTIVE ZEROS

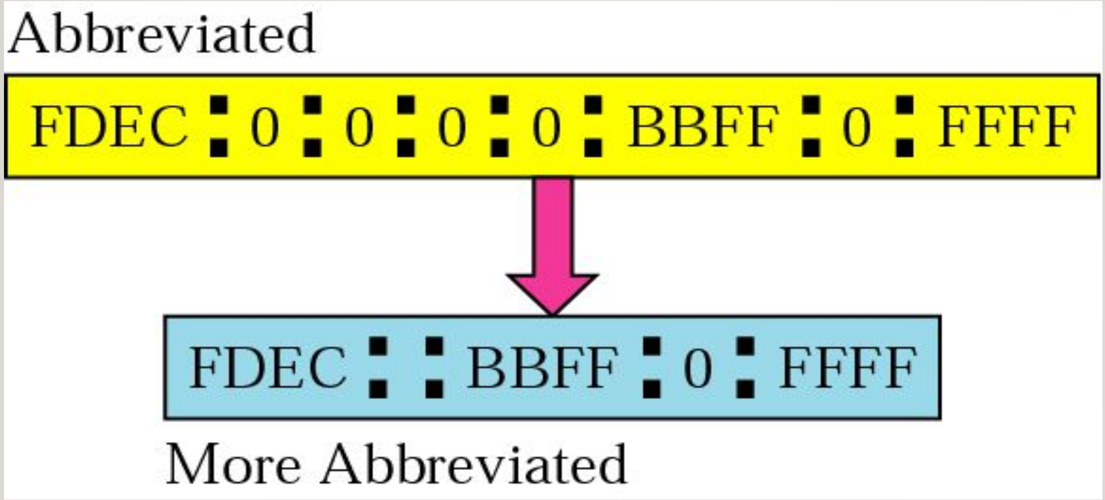


Figure IPv6 Abbreviated Address with Consecutive Zeros happy

LEARNING